



OCR and Machine Translation Pipeline for Low-Resource Sámi Languages



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1. Background & Motivation

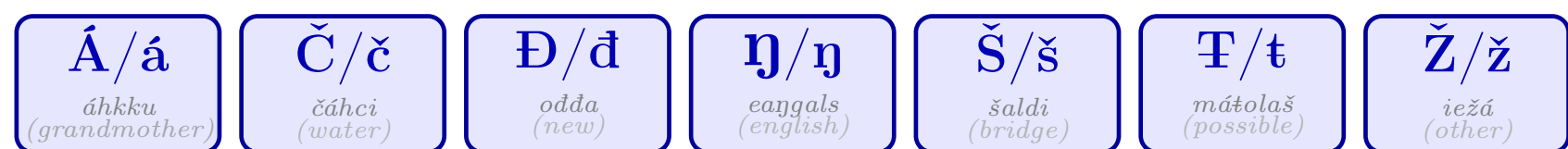
Sámi languages are a group of nine low-resource languages that belong to the family of Uralic languages. They are spoken in the Sápmi region, across Norway, Sweden, Finland, and Russia. North Sámi, the most widely spoken, has approximately 25,000 speakers, yet it remains classified as endangered by UNESCO.

Challenges in Optical Character Recognition (OCR) for North Sámi range from scarce training data to complex orthographic features, making it a difficult task to streamline the digitalization efforts. Additionally, the language characteristics of Sámi lead to a large number of inflected word forms, further complicating accurate recognition. While state-of-the-art models like TrOCR achieve high accuracy (91.56%), their computational demands (~350MB, ~12.41 s/img inference on CPU) make them ill-fitted for deployment in resource-constrained environments.

Objective was to develop and benchmark a lightweight CNN-CTC OCR model for Northern Sámi, evaluating its performance against the TrOCR transformer-based models and in different contexts such as in OCR→MT translation pipeline, out-of-domain generalization, linguistically motivated optimisation and applicability in sustainable AI.

okta lea dáhpáhus go bodii dárrolaš njuorjovuopmegierragii

It once happened that a Norwegian settler came to the uplands



Seven diacritical characters distinguish Sámi from standard Latin alphabet

(a) Example sentence from Johan Turi's *Muitalus sámiiid birra* (1910). (b) Seven Sámi-specific diacritics that distinguish the orthography.

2. Data, Methods & Metrics

Training data was Språkbanken synthetic Sámi OCR dataset (National Library of Norway), ~307k line images (276k training + 31k hyperparameter tuning), 32×800px.

Evaluation data used 1048 samples: 1000 in-domain synthetic lines (Språkbanken validation split) + 48 parallel Northern Sámi–English sentences from Turi's 1910 "An Account of Sámi".

Pipeline - a lightweight, modular OCR→MT:

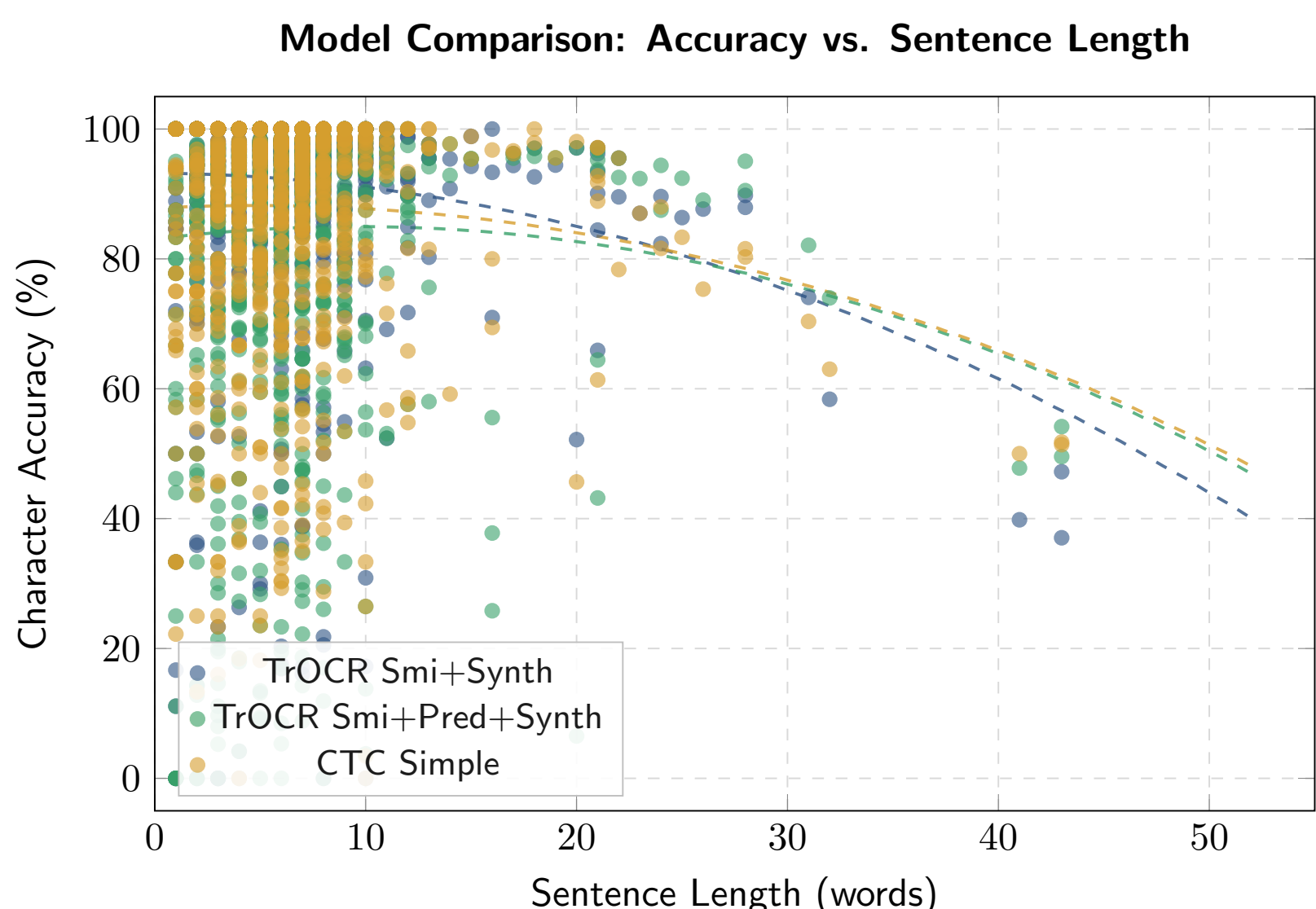
- Image preprocessing (grayscale, 32×800 px)
- 7-layer CNN backbone (64→128→256²→512³ ch.)
- Adaptive pool + linear projection to 256 dimensions
- CTC decoder over a 395-symbol vocabulary
- Translation via TartuNLP API

Training. AdamW (lr 10⁻⁴), batch 32, 100 epochs with early stopping, CTC loss, trained on a single NVIDIA V100. Inference benchmark on AMD Ryzen 5 PRO CPU.

Negative result. Pretrained VGG-16/19 and ResNet-50 backbones did not converge.

Metrics Used. **OCR:** Character Error Rate (CER), Word Error Rate (WER); Word Acc. = 100−WER, Char Acc. measured directly ($\approx 100 - \text{CER}$).

Accuracy vs. Sentence Length



Character accuracy vs. sentence length.
The character accuracy drops for longer sentences due to the fixed 32×800 px input.

3.1 Results: Benchmark

Model	Char. Acc. (%)	Word Acc. (%)	s/img	Size
trocr_smi_synth*	91.56	71.84	12.41	350MB
ctc_simple (ours)	87.72	62.65	0.033	23MB
trocr_smi_pred_synth*	84.15	54.24	12.08	350MB
trocr_smi	75.88	36.50	16.23	350MB

*smi=Sámi data, pred=+auto-transcribed, synth=+synthetic pre-training.

CNN-CTC achieves 87.72% char. acc.(12.24% CER).

Example predictions - ground truth, and error types.

Correct Recognition (CER = 0%)

mijáv juohkkaláhkáj kreatijvvan.

Ground Truth: mijáv juohkkaláhkáj kreatijvvan.

Prediction: mijáv juohkkaláhkáj kreatijvvan.

Diacritical Errors (CER = 8.3%)

oddasis ealáskahttojuvvon) Suohkana hálddahuslaš

Ground Truth: oddasis ealáskahttojuvvon) Suohkana hálddahuslaš

Prediction: oddasis ealáskahttojuvvon) Duohkama hálddahuslaš

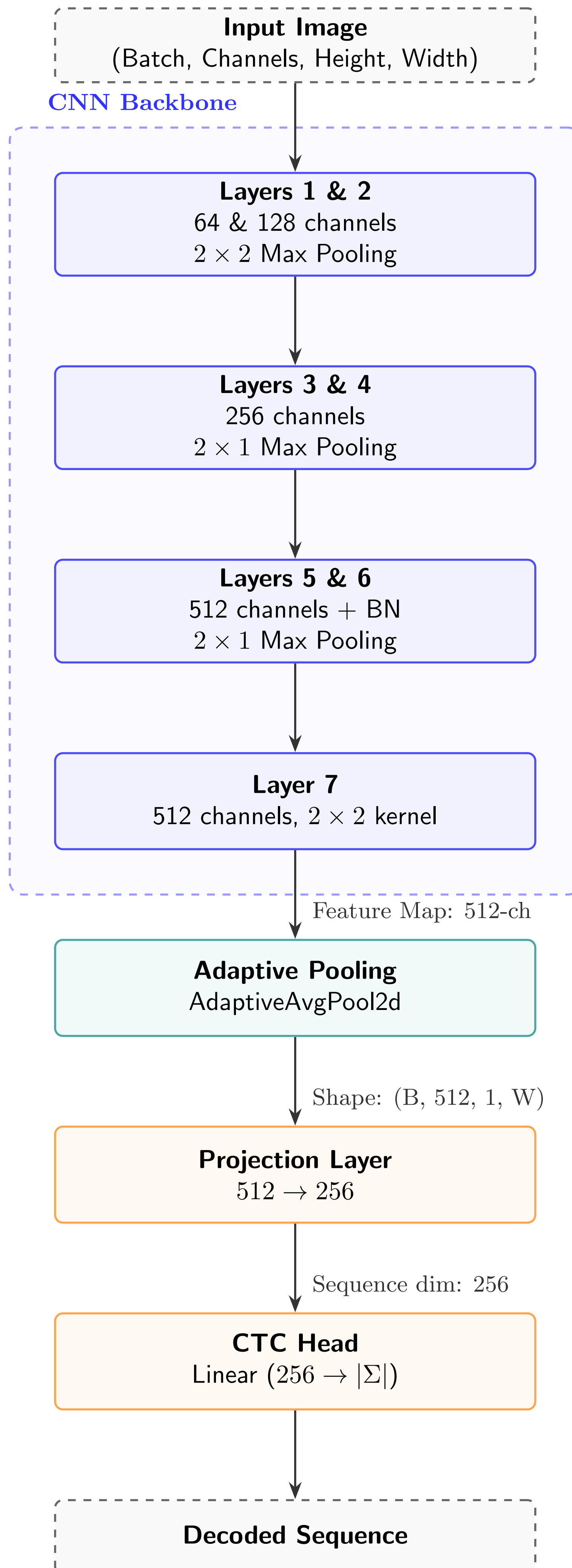
Uppercase Text Errors (CER = 29.6%)

FYLKKAGIELDDAIGUIN. FINNMÁRKKU FYLKKAGIELDDAS LEA ÁIGUMUŠŠIEHTADUS SÁMI

Ground Truth: FYLKKAGIELDDAIGUIN. FINNMÁRKKU
FYLKKAGIELDDAS LEA ÁIGUMUŠŠIEHTADUS SÁMI

Prediction: FYLKKAGIELDDAIGUI0. F100mÁRKKU
FYLKKAGIELDDAS LEA ÁIGUmUŠe

Architecture



CNN-CTC: 32×800 px input → CNN Backbone → Adaptive Pooling → Projection Layer → CTC head → Decoded Sequence.

3.2 Results: Other Findings

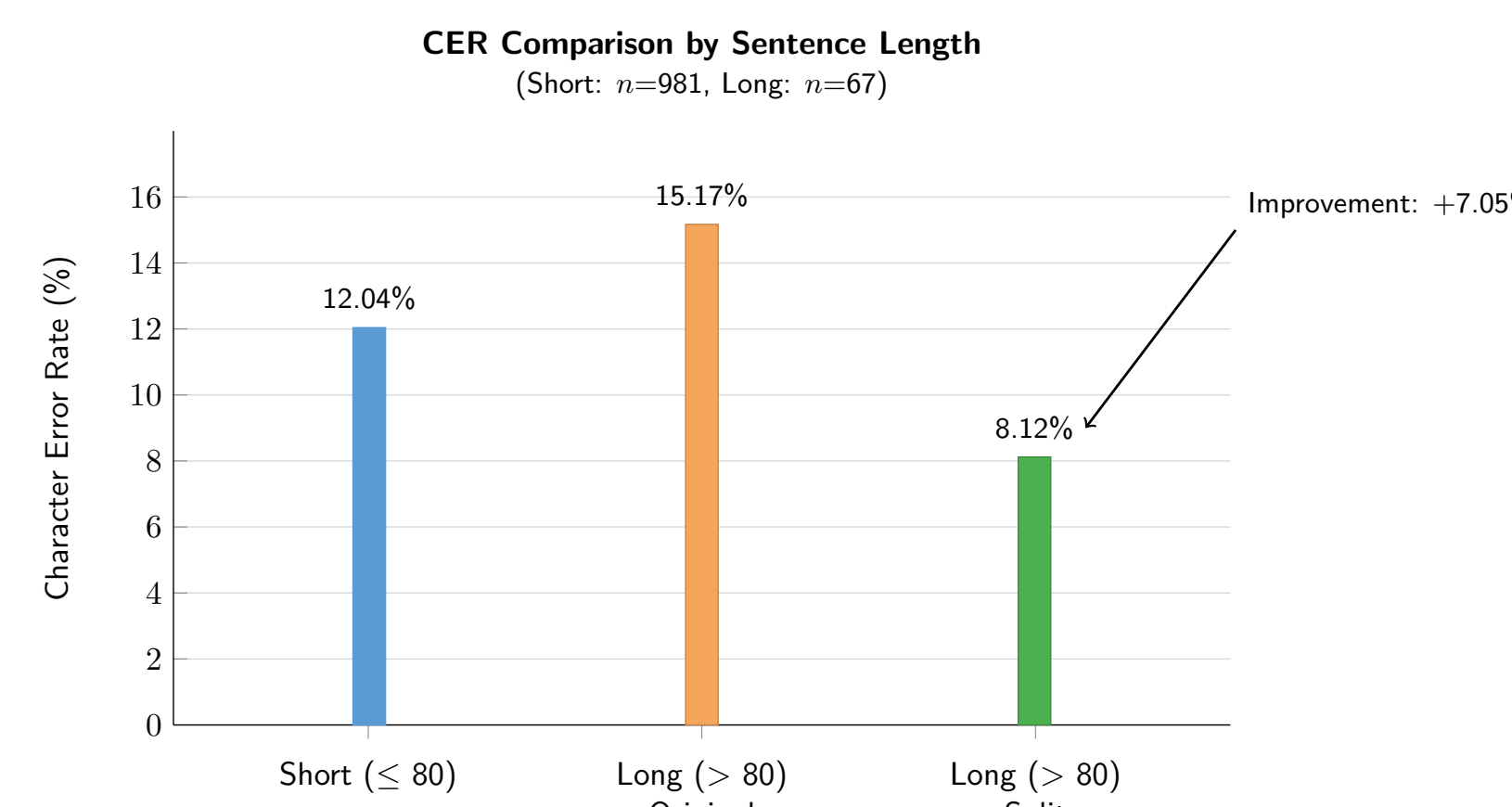
Generalization: in vs. out of domain.

Model	HF CER (%)	Book CER (%)	Δ (pp)
trocr_smi_synth	8.33	13.17	+4.84
ctc_simple (ours)	12.24	12.21	−0.03
trocr_smi_pred_synth	16.09	11.00	−5.09

HF = HuggingFace, in-domain. Book = out-of-domain historical text.

The best TrOCR degrades by 4.84 pp on historical text, while CNN-CTC is stable (12.24% → 12.21%).

Conjunction-based splitting on long sentences.



Linguistically motivated preprocessing splitting at Sámi conjunctions (*ja, muhto, go, ...*) cuts long-sentence CER from 15.17% to 8.12%.

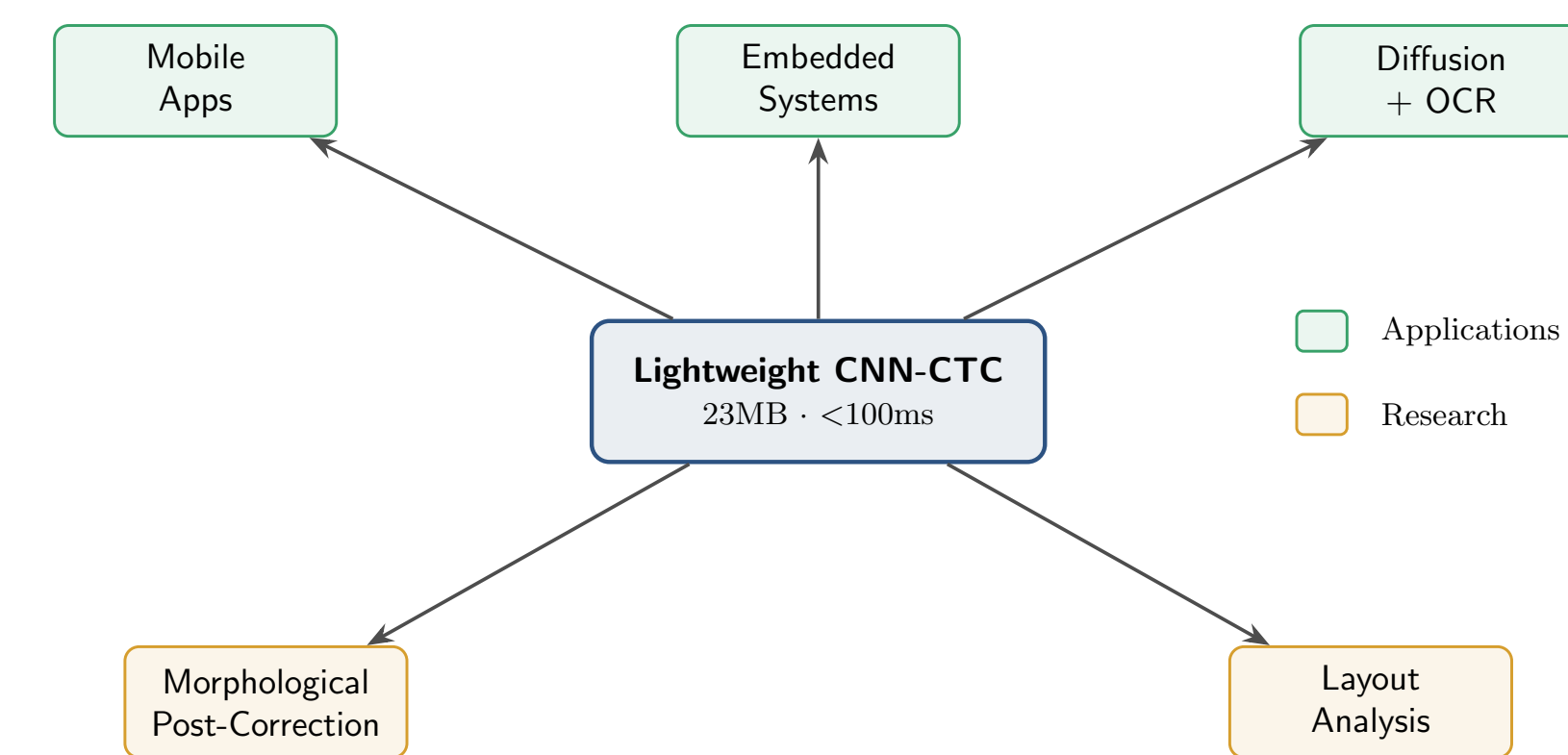
End-to-end OCR→MT. On 48 book sentences: BLEU 10.40, chrF 33.85, TER 76.30.

4. Conclusion & Future Work

Key Outcomes. The CNN-CTC achieves 87.72% character accuracy. The evaluation accuracy barely moves between synthetic and historical book text (12.24% → 12.21%). Yields an estimated throughput of ~109,000 img/h. The OCR→MT pipeline achieves BLEU 10.40 on the 48 parallel sentences and conjunction-based splitting recovers ~7 pp CER on long sentences.

Implications. The CNN-CTC model trails the best TrOCR by only 3.84pp character accuracy while being roughly 15× smaller and 380× faster, making it worthwhile for resource-constrained deployment. The in and out of domain accuracy stability lacks sufficient statistical power for concrete conclusions but it might suggest better lightweight generalisation. The MT quality indicates the need for improved English translation models for Sámi. Conjunction-based splitting proved to be a valid preprocessing proof-of-concept.

Future Work: Use real scanned documents, layout analysis for full-page processing, morphological post-correction, mobile / edge deployment for field digitization



Future work: real scans, layout analysis, morphological post-correction, mobile deployment.

References & Acknowledgments

Key references. [1] Enstad et al. *Comparative analysis of OCR for Sámi*, NoDaLiDa 2025. [2] Li et al. *TrOCR*, AAAI 2023. [3] Shi et al. *CRNN*, IEEE TPAMI 2017. [4] Corallo & Varde, *OCR for Amazigh*, AAAI Workshops 2023. [5] Turi, *Muitalus sámiiid birra*, 1910. [6] TartuNLP MT API, 2025.

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Code: <https://github.com/magwrap/north-sami-ocr>

